「Cauchy - Riemann equation.

Let $f: G \to \mathbb{C}$ be a map on region G and let $u(x,y) = \mathrm{Re}\, f(x,y)$, $v(x,y) = \mathrm{Im}\, f(x,y)$.
Suppose that u and v have continuous partial derivatives. Then,

f is analytic if and only if u and v satisfy the Cauchy-Riemann equation.

Proof. Suppose that f is analytic. For $h \in \mathbb{R} \setminus \{0\}$,

$$\frac{f(z+h)-f(z)}{h} = \frac{u(x+h,y)-u(x,y)}{h} + i \cdot \frac{v(x+h,y)-v(x,y)}{h}$$

Letting $h \to 0$, then $f'(z) = \frac{\partial u}{\partial x}(x,y) + i \cdot \frac{\partial v}{\partial x}(x,y)$.

Meanwhile, for $h \in \mathbb{R} \setminus \{0\}$,

$$\frac{f(z+ih)-f(z)}{ih} = -i \cdot \frac{u(x,y+h)-u(x,y)}{h} + i \cdot \frac{v(x,y+h)-v(x,y)}{ih}$$

Letting $h \to 0$, then $f'(z) = -i \cdot \frac{\partial u}{\partial y}(x,y) + \frac{\partial v}{\partial y}(x,y)$

So, $u_x = v_y$ and $v_x = -u_y$. $\cdots (*)$ Not using continuously differentiable.

Conversely, u and v satisfy the condition (*), and u_x, u_y, v_x, v_y are continuous.
Let $z = x+iy \in G$ be given. Since G is open, for some $r>0$, $B_r(z) \subseteq G$.
For $h = s+it \in B_r(0)$,

$$u(x+s, y+t) - u(x,y) = [u(x+s, y+t) - u(x, y+t)] + [u(x, y+t) - u(x,y)]$$

By the Mean-Value Theorem for variables s and t, respectively, (*)

$$u(x+s, y+t) - u(x, y+t) = u_x(x+s_1, y+t)\cdot s \quad \text{for some } |s_1| < |s|$$

and

$$u(x, y+t) - u(x,y) = u_y(x, y+t_1)\cdot t \quad \text{for some } |t_1| < |t|$$

Let $\varphi(s,t) = [u(x+s, y+t) - u(x,y)] - [u_x(x,y)\cdot s + u_y(x,y)\cdot t]$. Now, (*) gives

$$\frac{\varphi(s,t)}{s+it} = \frac{s}{s+it}\cdot\left[u_x(x+s_1, y+t) - u_x(x,y)\right] + \frac{t}{s+it}\cdot\left[u_y(x, y+t_1) - u_y(x,y)\right]$$

But, $|s_1| < |s| \le |s+it|$ and $|t_1| < |t| \le |s+it|$, so letting $s+it \to 0$, then

$$\frac{\varphi(s,t)}{s+it} \to 0$$

Therefore, $u(x+s, y+t) - u(x,y) = u_x(x,y)s + u_y(x,y)t + \varphi(s,t)$

and similarly $v(x+s, y+t) - v(x,y) = v_x(x,y)s + v_y(x,y)t + \psi(s,t)$ where $\lim\limits_{s+it\to 0} \frac{\psi(s,t)}{s+it} = 0$.

Now, $$\frac{f(z+s+it) - f(z)}{s+it} = u_x(z) + iv_x(z) + \frac{\varphi(s,t) + \psi(s,t)}{s+it} \to u_x(z) + iv_x(z).$$

as $s+it \to 0$.

「Lemma. Let $f: G \to \mathbb{C}$ be a Holomorphic at $z_0 \in G$ where G is an open set.
Then, for the differential operators

$$\frac{\partial}{\partial z} \overset{\text{def}}{=} \frac{1}{2} \cdot \left(\frac{\partial}{\partial x} + \frac{1}{i} \frac{\partial}{\partial y} \right) \quad \text{and} \quad \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \cdot \left(\frac{\partial}{\partial x} - \frac{1}{i} \cdot \frac{\partial}{\partial y} \right),$$

$$\frac{\partial f}{\partial \bar{z}}(z_0) = 0 \quad \text{and} \quad f'(z_0) = \frac{\partial f}{\partial z}(z_0) = 2 \cdot \frac{\partial u}{\partial z}(z_0),$$

Proof. Denote $f(x+iy) = u(x+iy) + i \cdot v(x+iy)$, then

$$\frac{\partial f}{\partial \bar{z}} = \frac{1}{2} \cdot \left(\frac{\partial u}{\partial x} - \frac{1}{i} \cdot \frac{\partial u}{\partial y} \right) + \frac{1}{2} \cdot \left(i \cdot \frac{\partial v}{\partial x} - \frac{\partial v}{\partial y} \right)$$

$$= \frac{1}{2} \cdot \left[\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} + i \cdot \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y} \right) \right]$$

so, f satisfies the Cauchy-Riemann Condition iff $\frac{\partial f}{\partial \bar{z}} = 0$.

And, $$\frac{\partial f}{\partial z} = \frac{1}{2} \cdot \left(\frac{\partial u}{\partial x} + \frac{1}{i} \cdot \frac{\partial u}{\partial y} \right) + \frac{1}{2} \cdot \left(i \cdot \frac{\partial v}{\partial x} + \frac{\partial v}{\partial y} \right)$$

$$= \frac{1}{2} \cdot \left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right) + \frac{i}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

⌈Polar-Coordinate form.

⌈Thm. Let $f = u + i \cdot v$ be a Holomorphic function on $\{r \cdot e^{i\theta} \mid r > 0, -\pi < \theta \leq \pi\}$,
Then, $u_r = \frac{1}{r} \cdot v_\theta$ and $u_\theta = -\frac{1}{r} \cdot v_r$.

Proof. By the Chain Rule with respect to $z = re^{i\theta}$, (and $\partial f / \partial \bar{z} = 0$, by Holomorphic)

$$f_r = \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial r} = f'(r e^{i\theta}) e^{i\theta} \quad \text{and} \quad f_\theta = \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial \theta} = f'(r \cdot e^{i\theta}) \cdot i r e^{i\theta}$$

Therefore, $f_r = \frac{1}{ir} f_\theta$, So $u_r + i \cdot v_r = \frac{1}{ir} \cdot u_\theta + \frac{1}{r} v_\theta = \frac{1}{r} v_\theta - i \cdot r \cdot u_\theta$.

Thus, $u_r = \frac{1}{r} v_\theta$ and $v_r = -r \cdot u_\theta$, ▨

⌈Using this, investigate $\operatorname{Log} z \overset{\text{def}}{=} \ln|z| + i \cdot \operatorname{Arg} z$.
Given non-zero Complex number $z \in \mathbb{C}$, $z = re^{i\theta}$ for some $r > 0$, $-\pi < \theta \leq \pi$. So,

$$\operatorname{Log} z = \ln r + i \cdot \theta.$$

Now, the partial derivatives becomes

$$\underbrace{\frac{\partial}{\partial r} \ln r = \frac{1}{r}}_{u_r} \quad \text{and} \quad \underbrace{\frac{\partial}{\partial r} \theta = 0}_{v_r} \quad \text{and} \quad \underbrace{\frac{\partial}{\partial \theta} \ln r = 0}_{u_\theta} \quad \text{and} \quad \underbrace{\frac{\partial}{\partial \theta} \theta = 1}_{v_\theta}.$$

Thus the $\operatorname{Log} z$ satisfies the Cauchy-Riemann equation on

$$D = \{re^{i\theta} \mid r > 0, -\pi < \theta < \pi\},$$

and the partial derivatives of $\ln r$ and θ are Continuous on D,
Consequently, $\operatorname{Log} z$ is Holomorphic on D.